How accurate LIGHT_ArF_ebeam is

The Ar/F_2 models v0_2 (0-D) and 1Dv0_2 (1-D), version 0.2.0-arf-ebeam, on 30 September 2026, on LIGHT's own input and under an electron beam, with their reaction rates

Bottom line. LIGHT_ArF_ebeam v0_2 reproduces the decays its rate constants came from [1, 2, 3] with an rms of 0.12 in the logarithm, and at the Electra laser [4] its own input gives 64 % of the measured formation efficiency $g_0 I_{sat}/P$ of 0.29.

It also gives 86 % of Keio's [5] at its printed pump and 138 % of Sandia's [6]. No rate was set by a laser, fluorescence or yield measurement, so these scores are predictions. The lever is the yield of ArF^* from Ar^* + F_2, measured at 0.5 [7] and the default. At the 1.0 that every published ArF model books, Electra's efficiency is 119 %. Rokni's pressure curve [8] is followed with an rms of 0.15 in the logarithm, and 1Dv0_2's centre cell gives v0_2's numbers to 4×10^{-4} on the own input and 1×10^{-4} under the beam.

Its largest misses are the VUV yield of pure argon, 215 % of Morozov's measurement [9], the Ar_2F^* fluorescence at high current, 155 to 167 % of Jancaitis's [2], and Rokni's fluorescence ratio for a $\times25$ change in current [8], 167 %. That last is the two-temperature tail table's, read below the ionization fractions it was checked at. On a Maxwellian EEDF the ratio is 96 %.

Under the electron-beam drive this copy adds, built from REBUILD's beam physics with the measured secondary spectrum [10], the same network gives 84 % of Electra's formation efficiency, 114 % of Keio's and 182 % of Sandia's, and Rokni's current ratio at 97 %, the own input's failure at low pumps gone. Pure argon improves to 175 % of Morozov's measurement and no further. That is the network's one-photon ceiling, 0.538 of the input, near the 0.53 Morozov quotes as the theoretical maximum [9, 11]. His argon lies at 62 % of that maximum, where he finds his krypton and xenon close to theirs.

[1] Chen and Payne, IEEE J. Quantum Electron. 15, 149 (1979). [2] Jancaitis, Lawrence Livermore report UCRL-53465 (1983). [3] Ueda, Kanada, Kitagawa and Takuma, Rev. Laser Eng. 11, 576 (1983), and Ueda et al., Rev. Laser Eng. 12, 357 (1984). [4] Wolford et al., High Energy Density Phys. 36, 100801 (2020). [5] Lee, Matsui, Kannari and Obara, IEEE Trans. Electron Devices 36, 2053 (1989). [6] Adams and Dillon, J. Appl. Phys. 70, 4073 (1991), abstract. [7] Setser, ERDA report COO-2807-11 (1977), OSTI 6681231. [8] Rokni, Jacob and Mangano, Phys. Rev. A 16, 2216 (1977). [9] Morozov, Heindl, Krücken, Ulrich and Wieser, J. Appl. Phys. 103, 103301 (2008). [10] Opal, Peterson and Beaty, J. Chem. Phys. 55, 4100 (1971). [11] Lorents, Physica 82C, 19 (1976), the maximum as quoted in [9].

LIGHT_ArF_ebeam is LIGHT_ArF, LIGHT converted to argon fluoride, with an electron-beam drive added to both its models. v0_2 is one well-mixed box of argon and fluorine, and 1Dv0_2 runs the same chemistry in every cell of a 1 cm gap on Nathaniel's Fluid1D. A given power is deposited uniformly, straight into the electrons as LIGHT's own input puts it, or by an electron beam, and neither model has a cavity. The basic network has 69 rows. Every rate cites a measurement whose page was read first-hand, many of them in review tables that quote the measurement, or is a band. A measured rate known only as a published spread keeps its measurement as the band's default, and a rate with no measurement is 0 by default. No rate was set from a laser, fluorescence or yield measurement.

Scored against every ArF measurement found, v0_2 on its own input gives Electra's formation efficiency at $\times0.64$ the measured value, Keio's at $\times0.86$ and Sandia's at $\times1.38$. The largest lever is the yield of ArF^* from Ar^* + F_2. At Setser's measured 0.5 Electra is under-predicted, and at the 1.0 every published ArF model books it is $\times1.19$. Rokni's pressure curve is followed with an rms of 0.15 in the logarithm and Jancaitis's current scan with 0.28. Two misses stand out. The network has no loss of Ar_2F^* that grows with the excitation, and pure argon gives $\times2.15$ Morozov's VUV efficiency. At low pumps LIGHT's own input runs hot, because the two-temperature tail it reads there is cold. On a Maxwellian EEDF the same input gives Rokni's current ratio at $\times0.96$, where the tail gives $\times1.67$. The 1-D model's centre cell gives the 0-D numbers to 4×10^{-4} at every device point on the own input and to 1×10^{-4} under the beam. Table 1 lists every test and Figure 1 shows them together.

test	measured	own input	own input, Maxwellian	beam drive	1Dv0_2 centre	role
Formation efficiency $g_0 I_{sat}/P$ and gain at the lasers, model over measured						
Electra at 1.09 MW/cm^3, Wolford 2020	0.29	×0.64	×0.71	×0.84	×0.64	held out
the same at 0.90 MW/cm^3, Petrova 2022	0.36	×0.55	×0.61	×0.72	×0.55	held out
ArF* yield of Ar* + F_2 at 1.0		×1.19		×1.08		sensitivity
Keio at its printed 0.51 MW/cm^3, Lee 1989	0.26	×0.86	×0.98	×1.14	×0.86	held out
NRL's replot at 1.21 MW/cm^3	0.11	×1.97	×2.20	×2.56	×1.97	held out
Sandia at 5.00 MW/cm^3, abstract only	0.16	×1.38	×1.55	×1.82	×1.38	held out
Electra gain at 1.36 over 0.82 atm	0.70	×1.31		×1.32	×1.31	held out
Electra gain for a pump ratio of 2.76	1.74	×1.36		×1.13		held out
Relative fluorescence, geometric mean of model over measured and the rms of its logarithm						
Rokni 1977 Fig. 4, 193 nm against Ar		0.97 (0.15)	1.05 (0.07)	0.99 (0.10)	1.00 (0.24)	held out
Rokni 1977 Fig. 5, ratio for ×25 in current	24 to 27	1.67 (0.58)	0.96 (0.09)	0.97 (0.05)		held out
Jancaitis 1983 Fig. 3, peaks against current		1.06 (0.28)	1.18 (0.32)	1.17 (0.35)		held out
Photons per initial excitation, and pure argon						
SRI 1976 Fig. 4, 193 nm against Ar		0.66 (0.47)				held out
SRI 1976 Fig. 4, 290 nm against Ar		0.90 (0.54)				held out
SRI 1976 Fig. 11, 193 nm against F_2		0.84 (0.24)				held out
Morozov 2008, pure-Ar VUV efficiency	0.33 ± 0.04	2.15 (0.77)		1.75 (0.56)		held out
The sources of the network's constants						
decays of Ar*, ArF* and Ar_2F^*, 69 points		0.97 (0.12)				consistency
Stern-Volmer plots, 15 points		1.06 (0.17)				consistency

Table 1: LIGHT_ArF_ebeam against the ArF data, every band at its default unless a row says otherwise. The own input is LIGHT's, the Maxwellian column is the same input without the two-temperature tail, and the beam column is the model's electron-beam drive. The 1Dv0_2 column is the 1-D centre cell on the own input. A × factor is model over measured.

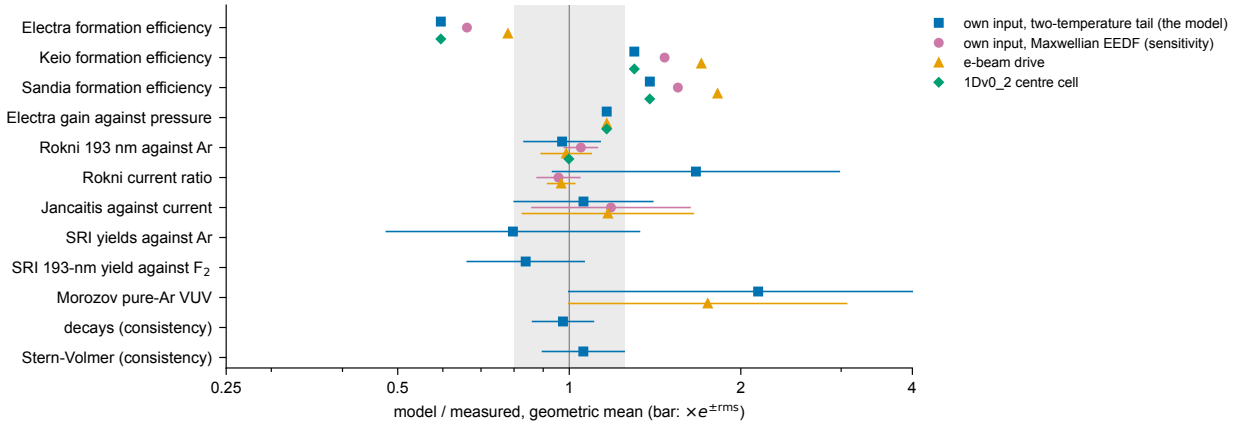


Figure 1: Every test at a glance, model over measured on a logarithmic axis. A bar spans the geometric mean times $e^{\pm \mathrm{rms}}$ where a test has three points or more. The grey band is 0.8 to 1.25.

1 How the numbers were made

Each dataset has one role. The decays and Stern-Volmer plots that the network's constants were read from check that the network reproduces them. Every other measurement is held out and scores the model. No dataset sets a rate, and no band is narrowed or moved by a score. This is the repository's predictive-only policy. A score is the ratio of model to measurement. For a set it is the geometric mean with the rms of the natural logarithms, and a relative curve leaves out its normalisation point.

Each test runs v0_2 three ways. LIGHT's own input puts the pump into the electron gas, and the two-temperature closure reads a tail table made from beam-sustained kinetic solutions. This is the model, and every score is its score unless a column says otherwise. The same input on a Maxwellian EEDF shows how much of a result the tail table makes. The electron-beam drive in `v0_2/beam.py` deposits the same power as ion pairs, excited atoms and secondary electrons. It runs REBUILD's Spencer-Fano cascade on the model's own cross sections with the measured secondary spectrum of Opal, Peterson and Beaty, $\bar{E}$ = 10.0 eV for argon, and gives 25.2 eV per ion pair and 0.39 excited atoms per pair at the reference fill, with 4.4 eV per pair to the electrons. W is an output, scored against the measured 26.4 eV and 4.5 % under it in pure argon. REBUILD's own rule, equal sharing read at one source energy, gives 30.85 eV and 0.84, a point on an oscillation that the measured spectrum removes. The seeded starts of SRI and Chen and Payne take the drive's excited atoms per ion. 1Dv0_2 runs both inputs on 64 cells, the beam's products added in every cell of its stage.

Every number in this note is written by `make_overview.py` from the records of `arf_data/score_arf.py`, made with the chemistry this note describes, and from fresh runs. `v0_2/tests/test_docs.py` fails when the generated files no longer match the network or the scores.

2 The ArF network

v0_2 keeps KrF v0_2's equations, energy ledger, two-temperature closure and bands. Its species, cross sections, tail table and rates are ArF's. Five things differ from KrF.

- One excimer state. No ArF C position, B–C mixing rate or equilibrium is measured, and the calculated B–C splitting is about 0.25 eV. ArF^* is ArF(B) alone.
- Stern-Volmer products. The ArF(B) radiative lifetime has never been measured, and its 4.2 ns is Dunning and Hay's calculation. The quenching of ArF(B) by F_2, Ar, 2Ar and electrons is measured as products $k\tau_B$, and the rows divide them by τ_B. A change of the calculated lifetime keeps every measured product.
- The harpoon yield. Ar^* + F_2 makes ArF^* with yield 0.5, Setser's flowing-afterglow measurement, where every published ArF model books 1.0.
- The argon levels. Each of the 37 excitation blocks of the IST-Lisbon set is its own row. The 4 4s blocks make Ar^*, and the 33 higher ones cascade to it at once.
- Ion-ion recombination. No absolute measurement exists. The rows use Hoffman and Moreno's calculation as Lee 1989 prints it, with the Debye screening and the Thomson $(T_g/300)^{-3}$ that KrF's rows carry.

The network carries 14 bands, 5 of them for rates with no measurement. Appendix A lists every row, band and species energy.

3 The reference point

At the reference point, 1 atm with 0.5 % F_2 and 300 kW/cm^3 for 1 μs followed to 2 μs, v0_2 radiates 5.05 % of the deposited energy at 193 nm (Table 2). The ArF(B) formed carries 18.7 % of the input as 193-nm quanta. It comes 83 % from Ar^* + F_2, 13 % from Ar_2^* + F_2 and 4 % from ion-ion recombination.

Of the ArF(B) formed, 27 % radiates, 36 % forms Ar_2F^*, 24 % is quenched by Ar and 11 % by F_2. Of the Ar^* formed, 33 % makes ArF(B), 33 % reacts with F_2 without making it and 33 % forms Ar_2^*. The F_2 falls below a tenth of its fill at 728 ns. T_e averages 6.8 eV over the pulse and T_g ends at 675 K. The ledger closes to 4×10^{-14} of the input.

On a Maxwellian EEDF the same point gives 5.89 % at 193 nm at a mean T_e of 1.8 eV. The seed density sets only the first nanoseconds, and seeds of 1.0×10^{10} to 1.0×10^{15} cm^{-3} give 5.05 to 5.06 %.

where the deposited energy is at the end	% of the input
193 nm (ArF B–X)	5.05
284 nm (Ar_2F^*)	0.67
127 nm (Ar_2^*)	4.60
heat, elastic	8.87
heat, reactions and captured electrons	54.85
stored, ions (Ar^+, Ar_2^+, F-)	−0.84
stored, Ar^*	0.41
stored, Ar_2^*	15.93
stored, ArF(B)	0.00
stored, Ar_2F^*	0.00
stored, F atoms (F_2 broken)	10.45
stored, electron gas	0.00

Table 2: The reference point's energy at 2 μs. The stored entries are formation energy still held. The ledger closes to 4×10^{-14}.

4 The bands

Each band at either end, the others at their defaults, gives Table 3. Three rates are the levers at the reference point. Ar_2^* + F, which has no measurement, takes the 193-nm share from 5.05 to 8.56 % at the top of its band. The ArF* yield of Ar^* + F_2 spans 3.01 to 8.89 %. The trimer's $k\tau_B$ spans 3.56 to 5.63 %, and the trimer's gas temperature law at the bottom of its band gives 7.01 %. Ar^* + 2Ar spans 4.90 to 5.89 %, and every other band moves the 193-nm share by 0.28 percentage points or less.

band	low end	193 nm, %	127 nm, %	high end	193 nm, %	127 nm, %
ar_f2	6.4×10^{-10}	5.12	4.6	1.1×10^{-9}	4.99	4.6
arf_yield	0.25	3.01	4.3	1.00	8.89	5.2
ar2_form	2.8×10^{-33}	4.90	3.4	3.3×10^{-32}	5.89	5.9
ar2_rad	3.1×10^{5}	5.05	4.6	6.0×10^{7}	4.82	19.1
ar2_f2	2.5×10^{-10}	4.99	4.7	5.2×10^{-10}	5.05	4.6
ar2_f2_arf	0.62	4.84	4.6	1.00	5.12	4.6
ar2_f	0	5.05	4.6	1.0×10^{-9}	8.56	0.0
ar2_pen	0	5.05	4.6	5.0×10^{-10}	5.00	1.4
ar_pen	0	5.05	4.6	6.2×10^{-10}	4.77	2.5
arf_trimer	1.6×10^{-39}	5.63	4.7	5.0×10^{-39}	3.56	4.4
trimer_tg_pow	−3.90	7.01	4.8	0.00	5.05	4.6
arf_e	4.0×10^{-16}	5.08	4.6	7.5×10^{-16}	5.02	4.6
ar2f_f	9.7×10^{-11}	5.05	4.6	1.9×10^{-10}	5.05	4.6
ar2f_ar	0	5.05	4.6	2.2×10^{-14}	5.05	4.6

Table 3: The reference point with one band at a time at either end of its range. At the defaults 193 nm takes 5.05 % and 127 nm 4.60 %. Appendix A.3 gives each band's source and units.

5 Formation efficiency and gain at the lasers

A published small-signal gain g_0 and saturation intensity I_{sat} give a formation efficiency that needs no cross section, $g_0 I_{sat}/P$. It is the ArF(B) formation rate times the 193-nm quantum over the deposited power at the quasi-steady peak, and v0_2 makes the same quantity from its formation rate in the pulse's window (Figure 2).

At Electra's 1.09 MW/cm^3, the measured 0.29 is met at $\times$0.64 on the own input and $\times$0.84 under the beam drive. With the ArF* yield at 1.0 the own input gives $\times$1.19 and the beam drive $\times$1.08, so the measured yield of 0.5 and the published models' 1.0 bracket Electra. Keio's printed pump gives $\times$0.86 and Sandia's abstract $\times$1.38. NRL replots Keio's point at 1.21 MW/cm^3, which would make it $\times$1.97, so Keio's pump reading decides whether v0_2 is low or high there.

Electra's single-pass gain falls to 0.70 of its 0.82-atm value at 1.36 atm, and v0_2's falls to 0.92 (Table 4). Wolford notes that Electra's beam is not uniform at low pressure. The gain itself needs the ArF(B) stimulated emission cross section, which is only calculated. With Lee's value v0_2's g_0 is in the last column.

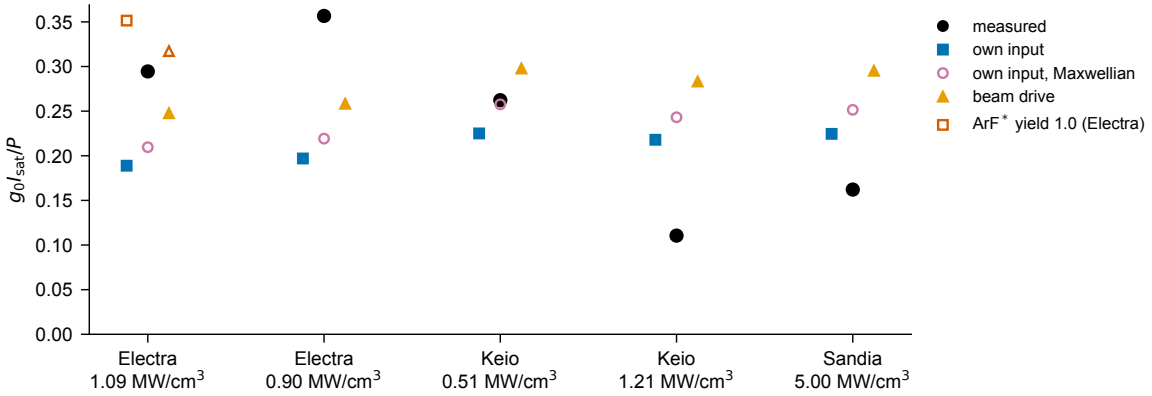


Figure 2: Formation efficiency $g_0 I_{sat}/P$ at the three lasers. Open symbols are the own input on a Maxwellian EEDF and, at Electra's 1.09 MW/cm^3, the ArF* yield at 1.0.

atm	g_0 measured, %/cm	measured	own input	beam drive	1Dv0_2 centre	g_0 with Lee's σ, %/cm
0.82	1.84	1.00	1.00	1.00	1.00	3.04
1.09	1.77	0.96	0.99	0.99	0.99	3.02
1.36	1.29	0.70	0.92	0.92	0.92	2.80

Table 4: Electra's single-pass gain against pressure, each column relative to its value at 0.82 atm. The model's ratios are its ArF(B) density ratios and need no cross section.

6 Fluorescence against pressure and current

Rokni, Jacob and Mangano measured the steady 193-nm fluorescence under a 5-A/cm^2 beam with 2 Torr of F_2 from 0.24 to 4 atm of argon (Figure 3). Normalised at 1 atm, v0_2 follows it with an rms of 0.15 in the logarithm on its own input and 0.10 under the beam drive. The deposition is taken as 20 kW/cm^3 per atm, an estimate, which sets the regime of the own input and not the normalised curve. In the same paper a current changed $\times$25 changes the fluorescence by 24 to 27, the ion channel forming ArF* with branching near one. The beam drive gives 0.97 of that, and the own input 1.67 (Table 5). Section 10 gives the reason.

Jancaitis and Powell measured the peak ArF* and Ar_2F^* fluorescence against the beam's current density

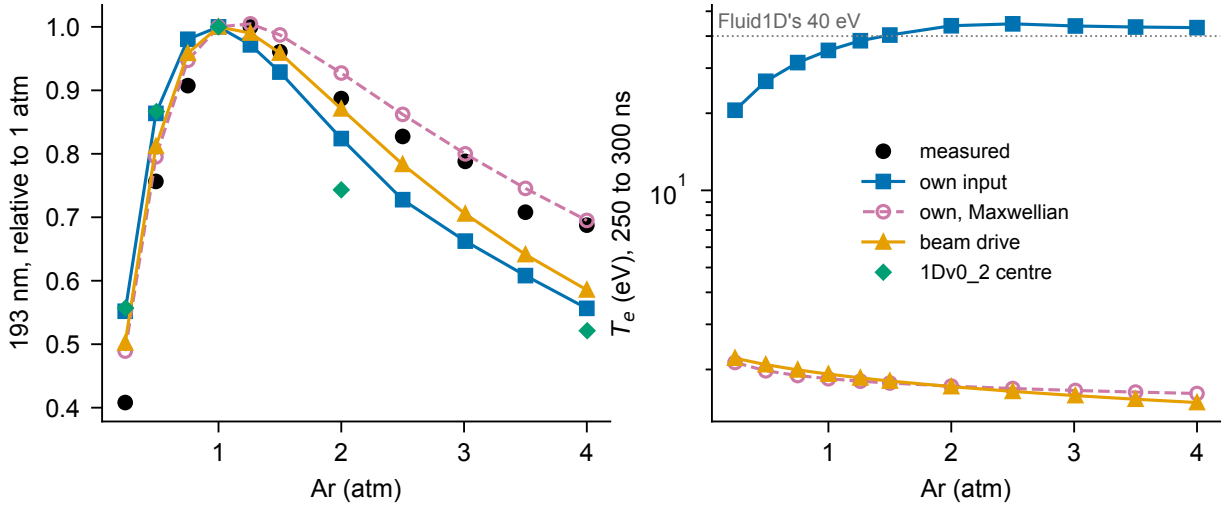


Figure 3: Rokni 1977 Fig. 4. Left, the 193-nm fluorescence against argon pressure, relative to 1 atm. Right, the model's T_e over the last 50 ns of the 300-ns pulse.

		own input			own, Maxwellian		beam drive	
Ar, atm	measured	ratio	T_e, eV	T_e at 1/25	ratio	T_e, eV	ratio	T_e, eV
0.26	25.6	27.0	21	94	21.9	2.1	23.6	2.2
1.01	26.6	50.6	35	71	25.0	1.8	24.8	1.9
2.02	24.5	52.1	44	55	25.0	1.7	24.7	1.7
3.53	24.3	44.1	43	40	24.9	1.6	24.6	1.5

Table 5: Rokni 1977 Fig. 5, the 193-nm fluorescence ratio for a $\times 25$ change of the beam current, with the model's mean T_e over 250 to 300 ns at the full current, and at 1/25 of it for the own input.

from 4.3 to 562 A/cm^2 (Figure 4). v0_2's ArF* peak follows the measured one with an rms of 0.24 in the logarithm, and at 562 A/cm^2 it is $\times 0.72$. Its Ar_2F^* keeps rising where the measured one saturates, $\times 1.55$ and $\times 1.67$ at 430 and 540 A/cm^2. Jancaitis attributes the saturation to a loss of Ar_2F^* that grows with the excitation. He estimates it as $Ar_2F^* + e^-$, which is not measured, and the network does not carry it.

7 Photons per initial excitation

SRI measured the photons at 193 and 290 nm per ion and excited atom made by a 3-ns Febetron pulse, against argon and against F_2 (Figure 5). v0_2 starts from SRI's excitation and follows the afterglow. Its 193-nm yield is 0.66 of the measured against argon, 0.45 to 0.89 over the 6 points, and 0.84 against F_2, 0.67 to 1.28. The model makes too little ArF* per excitation, the same question as at Electra. Its 290-nm Ar_2F^* yield runs from $\times 0.35$ at 1 atm to $\times 1.38$ at 9 atm, too little Ar_2F^* at low pressure and too much at high pressure. At high pressure, as at Jancaitis's high currents, the network lacks a loss of Ar_2F^*.

8 Pure argon

Morozov and coworkers measured the VUV power of pure argon under a low-energy beam at 0.33 ± 0.04 of the deposited power, independent of pressure from 500 to 1400 hPa (Figure 6). Under the beam drive v0_2 turns each ion and excitation into one Ar_2^* photon, 0.55 of the input, and its own input gives 0.67. The measurement is 0.60 of the beam drive's value, and no band closes the gap. At 1000 hPa the

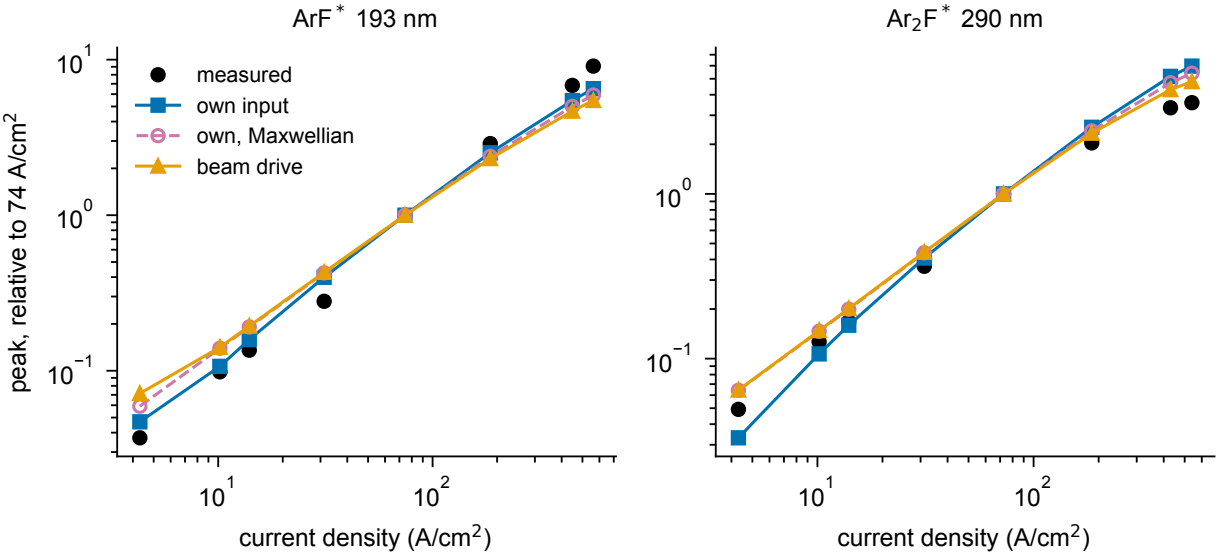


Figure 4: Jancaitis 1983 Fig. 3, peak fluorescence against current density, each curve relative to its point nearest 74 A/cm^2.

ends of the Ar* and Ar_2^* bands give 0.529 to 0.549 under the beam drive, the lowest with ar2_pen at its top. KrF's pure-krypton check misses the same way. The missing loss is not in the network. The drive's 25.2 eV per pair in pure argon puts the ceiling at 0.538, near the 0.53 Morozov quotes as the theoretical maximum. He finds his krypton and xenon close to their maxima and his argon at 62 % of it, so no input closes the gap. Either the network lacks an argon VUV loss that has been measured, or his argon value is low.

9 The sources of the constants

The network's constants are means of the groups that measured them. Run back against those measurements, v0_2 reproduces the decays with a geometric mean of 0.97 and an rms of 0.12 over 69 points, and the Stern-Volmer plots with 1.06 and 0.17 (Table 6). Rokni's F_2 plot, $\times 1.23$, is a steady e-beam run on the own input at a low pump, where Section 10 applies. These are checks of the bookkeeping, not of accuracy.

set	points	model over measured	rms of the logarithm
Chen and Payne Fig. 13, ArF* (follows Ar*)	9	0.97	0.11
Chen and Payne Fig. 14, Ar_2F^*	15	0.90	0.15
Jancaitis Fig. 7, Ar_2F^*, Ar/F_2 300	11	1.05	0.08
Jancaitis Fig. 7, Ar_2F^*, Ar/F_2 600	9	1.06	0.07
Ueda 1983 Fig. 8, Ar_2F^*	10	1.00	0.03
Ueda 1983 Fig. 12, Ar*(3P2) at 2.5 atm	5	0.85	0.20
Ueda 1983 Fig. 10, Ar*(3P2) in pure Ar	10	0.99	0.16
Chen and Payne Fig. 16, 1/I vs F_2 at 600 Torr Ar	4	0.88	0.13
Chen and Payne Fig. 17, 1/I vs Ar^2 at 0.84 Torr F_2	4	0.98	0.05
Rokni Fig. 3, 1/I vs F_2 at 150 Torr Ar	7	1.23	0.22

Table 6: The decays and Stern-Volmer plots the constants came from.

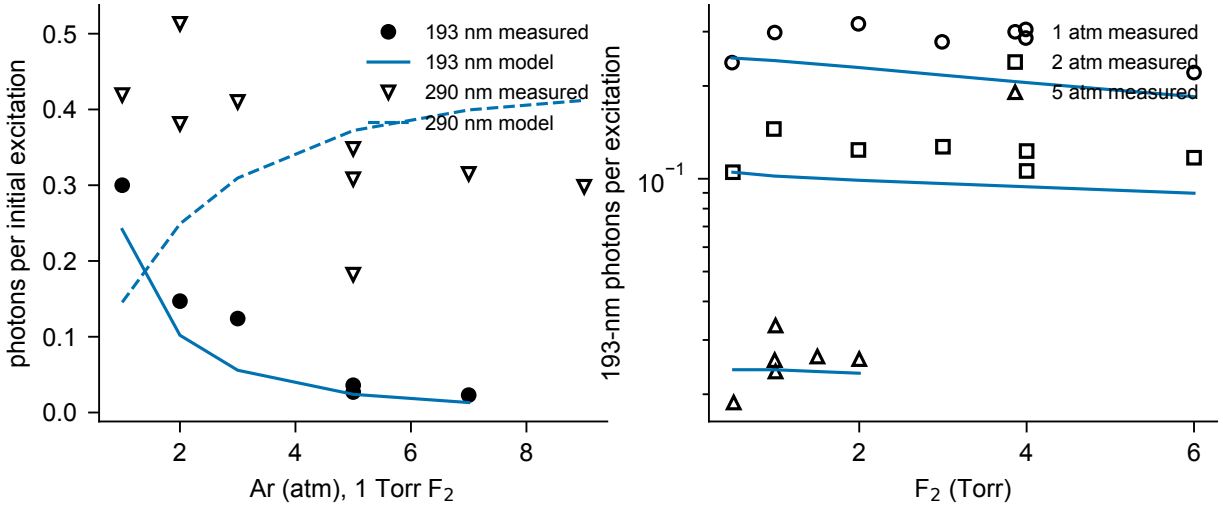


Figure 5: SRI MP-76-99. Left, Fig. 4, the yields against argon at 1 Torr of F_2. Right, Fig. 11, the 193-nm yield against F_2 at 1, 2 and 5 atm. Lines are v0_2.

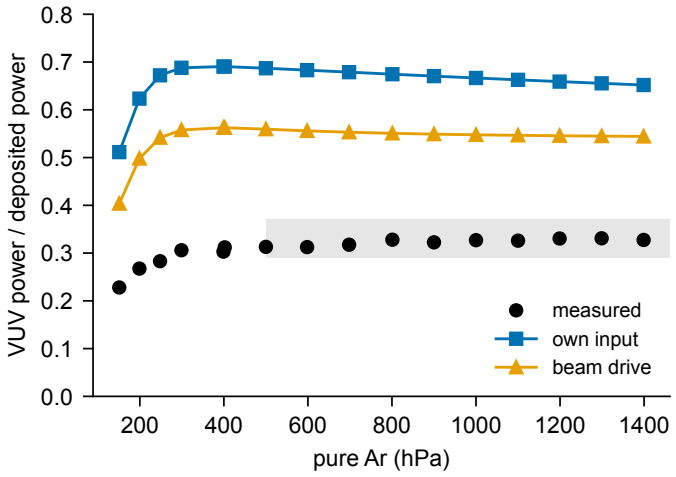


Figure 6: Morozov 2008 Fig. 10, pure-argon VUV power over deposited power. The grey band is the printed 0.33 ± 0.04.

10 The electron closure and the input

LIGHT's own input puts the pump into the electrons' bulk. The two-temperature closure gives the argon rows a tail above 11.548 eV at $T_{\text{tail}} = r\,T_e$, with r read from a table made from beam-sustained kinetic solutions. The table was checked against those solutions where n_e/N is at least 10^{-6}. Rokni's runs sit near $n_e/N = 2\times10^{-7}$, where the table gives $r = 0.010$. With so cold a tail the bulk has to reach 21 to 45 eV before the tail excites enough to take the pump. On a Maxwellian EEDF the same runs sit at 1.6 to 2.1 eV, where the beam drive sits at 1.5 to 2.2 eV (Table 7). The Maxwellian own input also follows Rokni's pressure curve more closely, $\times1.20$, $\times1.05$ and $\times1.01$ at 0.24, 2 and 4 atm against $\times1.35$, $\times0.93$ and $\times0.81$ with the tail, and it gives the ratio for $\times25$ in current at $\times0.96$ against $\times1.67$, as the beam drive's $\times0.97$ does. At the gain points the pump is higher, the tail warmer and the difference smaller. Electra goes from $\times0.64$ to $\times0.71$.

The own input's scores at low pumps are therefore the tail table's as much as the network's. The beam drive, whose secondaries carry the excitation, does not depend on it.

runs	own input	own input, Maxwellian	beam drive
Rokni Fig. 4, 20 kW/cm^3 per atm	20.5 to 44.7	1.6 to 2.1	1.5 to 2.2
Rokni Fig. 5, full current	21.0 to 44.0	1.6 to 2.1	1.5 to 2.2
Rokni Fig. 5, 1/25 of the current	39.6 to 94.2	1.6 to 2.1	1.5 to 2.2
Jancaitis Fig. 3, 20 to 50 ns	4.2 to 19.4	1.8 to 1.9	1.7 to 2.0
Electra, Keio, Sandia gain points	4.1 to 7.3	1.8 to 2.1	1.6 to 1.9

Table 7: The model's T_e in eV, over 250 to 300 ns at Rokni's conditions, 20 to 50 ns at Jancaitis's and in the formation window at the gain points.

11 The 1-D model

1Dv0_2 runs v0_2's chemistry, byte for byte, in each cell of Fluid1D's gap. At the device points its centre cell gives v0_2's numbers to 4×10^{-4}, the largest at Sandia, and its gap average lies 3.1 % lower on 64 cells. The difference is the wall cells, whose sheaths the mesh does not resolve. Table 8 gives the reference point on 64 and 128 cells. The centre cell agrees with v0_2 to 2×10^{-5} and 2×10^{-5}. In the wall cells the mean-energy bounds cut 3.11 and 1.55 % of the input, Fluid1D's split keeps out phantom negative field work of 12.7 and 2.4 %, and 0.040 and 0.036 % is lost to the walls. Under the beam the 1-D stage adds the beam's ion pairs, with their electrons, excited atoms and prompt F_2 dissociations, in every cell and substep, and its centre cell gives v0_2's beam numbers to 1×10^{-4}, the largest at Sandia, with the gap average 2.4 % lower. The own input's 1-D runs are unchanged by it, bit for bit.

	v0_2	64 cells		128 cells	
		centre	gap	centre	gap
193 nm, % of the input	5.0522	5.0523	4.8958	5.0523	4.9744
127 nm, % of the input	4.6040	4.6039	4.4585	4.6039	4.5295
T_g at the end, K	675.2	675.2	663.5	675.2	669.4

Table 8: The reference point in v0_2 and in 1Dv0_2 on two meshes.

Fluid1D holds the mean electron energy at or below 60 eV, a T_e of 40 eV. Where v0_2's T_e passes it, at Rokni's 2 and 4 atm with 44 and 43 eV, 1Dv0_2's centre gives $\times0.90$ and $\times0.94$ of v0_2. Under the beam v0_2's T_e stays at 1.5 to 2.2 eV over all of Rokni's pressures, and at these two the centre gives $\times1.00$ and $\times1.00$ of v0_2's beam run. Figure 7 follows the own input's 2-atm run. From 100 ns the centre cell and v0_2 agree to 0.3 % in every species but Ar^+ until their electron temperatures part by 1 % at 251 ns, as the 1-D T_e reaches the ceiling. Ar^+, whose production is steep in T_e, already differs by 7 % at that moment. After it v0_2's T_e climbs to 47 eV. Above the 30-eV edge of the rate tables the rows on Maxwellian tables are held, but the tail rows follow $T_{tail} = r\,T_e$, which stays far below the edge. At this state $r = 0.010$, and the ionization rate C2 is 3.2×10^{-17} at 30 eV, 1.8×10^{-15} at 40 eV and 1.3×10^{-14} cm^{3}s^{-1} at v0_2's 47 eV. v0_2's ionization keeps rising while the 1-D's is held, and by 300 ns v0_2 holds 8.0 times the 1-D's Ar^+. Holding the $Ar_2^+ + e^-$ power law at its 30-eV value in v0_2, as the 1-D stage does, moves the emission by 0.06 % or less.

12 The energy ladder

Figure 8 places every species of v0_2 at its formation energy, the energy above ground-state Ar and F_2 at rest that Appendix A lists, and draws the rows that move energy between them. It is LIGHT's ladder with v0_2's own numbers. ArF(B) lies 5.95 eV below the ion pair $Ar^+ + F^-$ and 6.41 eV above Ar + F, exactly the 193-nm quantum of 6.41 eV, so the photon leaves ArF(X) with 0.00 eV to heat. The harpoon G3 heats the gas by 3.53 eV for each ArF(B) it makes, $Ar_2^* + F_2$ through NK7 by 2.85 eV, the ion-ion recombinations K26 and K108 by 5.95 and 4.72 eV, and the trimer's formation by 0.65 eV. Ar_2F^* emits 4.37 eV at 284 nm and leaves 1.40 eV of lower-state repulsion, and Ar_2^* emits 9.76 eV at 127 nm and

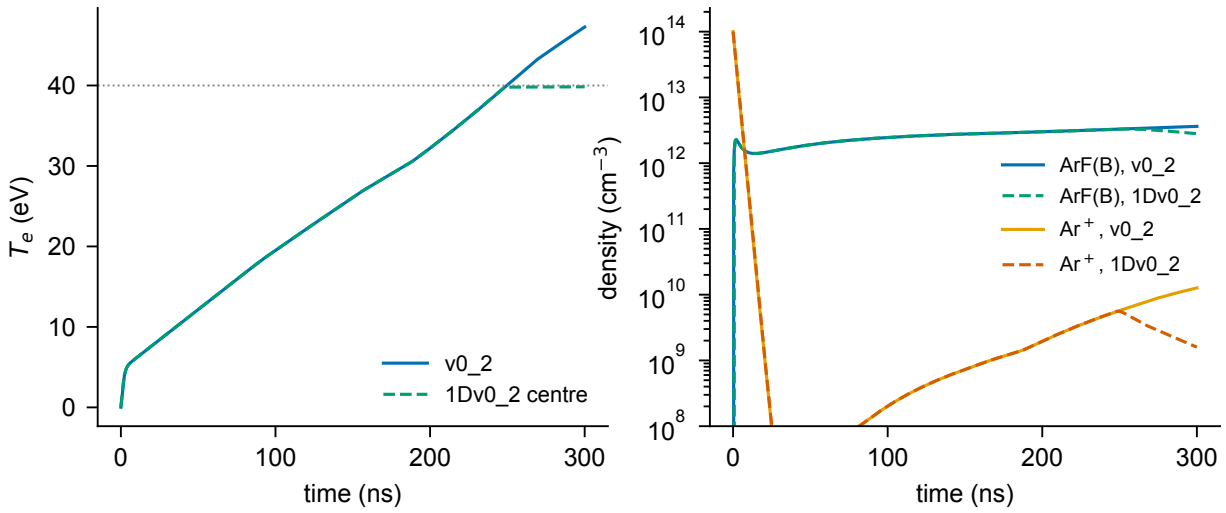


Figure 7: Rokni's 2-atm run in v0_2 and in 1Dv0_2's centre cell. Left, T_e and Fluid1D's ceiling. Right, ArF(B) and Ar^+.

leaves 1.11 eV. The 4 G1 rows excite the 4s levels, with thresholds from 11.548 to 11.828 eV, and the 33 C1 rows the higher levels, from 12.907 to 14.967 eV. Each family is drawn as one arrow.

13 The system map

The map on page 12 draws v0_2's basic network at the reference point, 1 atm of Ar with 0.5 % F_2, 300 kW cm^{-3} for 1 μs, with n_e seeded at 10^{15} cm^{-3} and then free and every band at its default. Band widths are proportional to the energy they carry, and energy is conserved at each of the map's 37 nodes to within 10^{-8} of E_{dep}. Of the deposited energy, 18.7 % passes through ArF(B) in 6.41-eV quanta. Of that ArF(B), 27.0 % radiates at 193 nm, 36.3 % forms the trimer and 23.7 % is quenched by argon, so 5.1 % of the deposited energy leaves at 193 nm, and 4.6 % leaves at 127 nm from Ar_2^*. A further 16.3 % is still held at 2 μs, most of it in Ar_2^*, whose radiative lifetime is 3.2 μs.

14 Where LIGHT_ArF_ebeam is not yet reliable

1. The ArF* made per excitation. Electra at $\times$0.64 on the own input and $\times$0.84 under the beam, SRI's 193-nm yield at 0.66 and the yield band that brackets Electra all point to it. The measured yield 0.5 stays the default. A measurement of the ArF* yield of Ar* + F_2 near 1 atm would settle it.
2. The loss of Ar_2F^*. The network has no Ar_2F^* + e$^-$ and no other loss that grows with excitation. Jancaitis's Ar_2F^* and SRI's 290-nm yield at high pressure call for one.
3. Pure argon. The VUV efficiency is 0.60 of the one-photon value, and no band closes it.
4. The electron closure at low pumps. The tail table is read below the ionization fractions it was checked at, and the own input's results there depend on it (Section 10). The beam drive does not.
5. The formation efficiency against pump. Under the beam the model makes ArF* at nearly the same share of the input at every device pump, where the measured share falls from Electra to Sandia, and Sandia rises to $\times$1.82.
6. Electra's gain against pressure, measured to fall faster than v0_2's.
7. What was never measured. The ArF(B) radiative lifetime, the ArF C state, Ar_2^* + F and the Penning rates are calculations or bands. F + F + Ar is known here only through Baulch 1981. Sandia's efficiency is from its abstract, and Keio's two pump readings give $\times$0.86 and $\times$1.97.

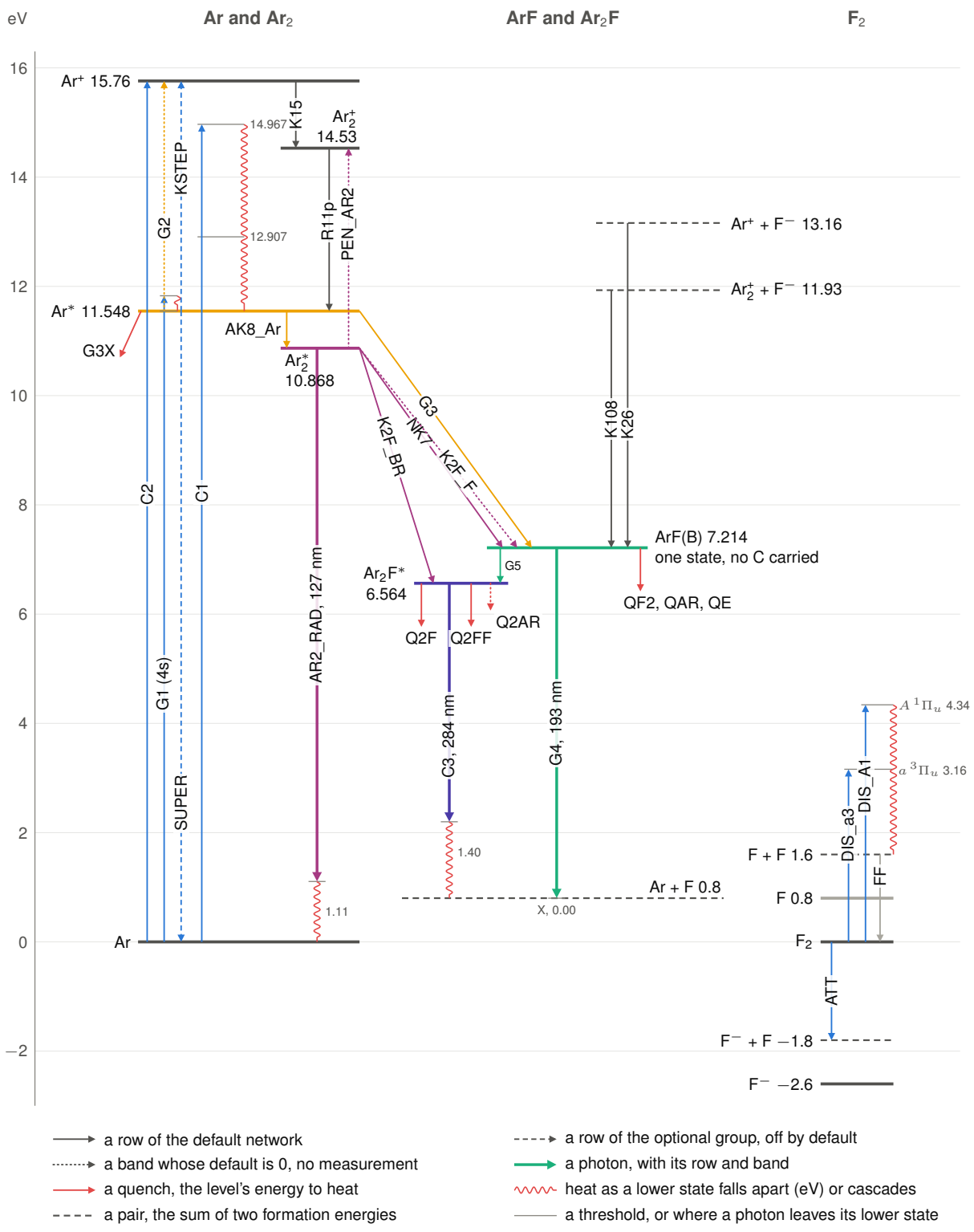
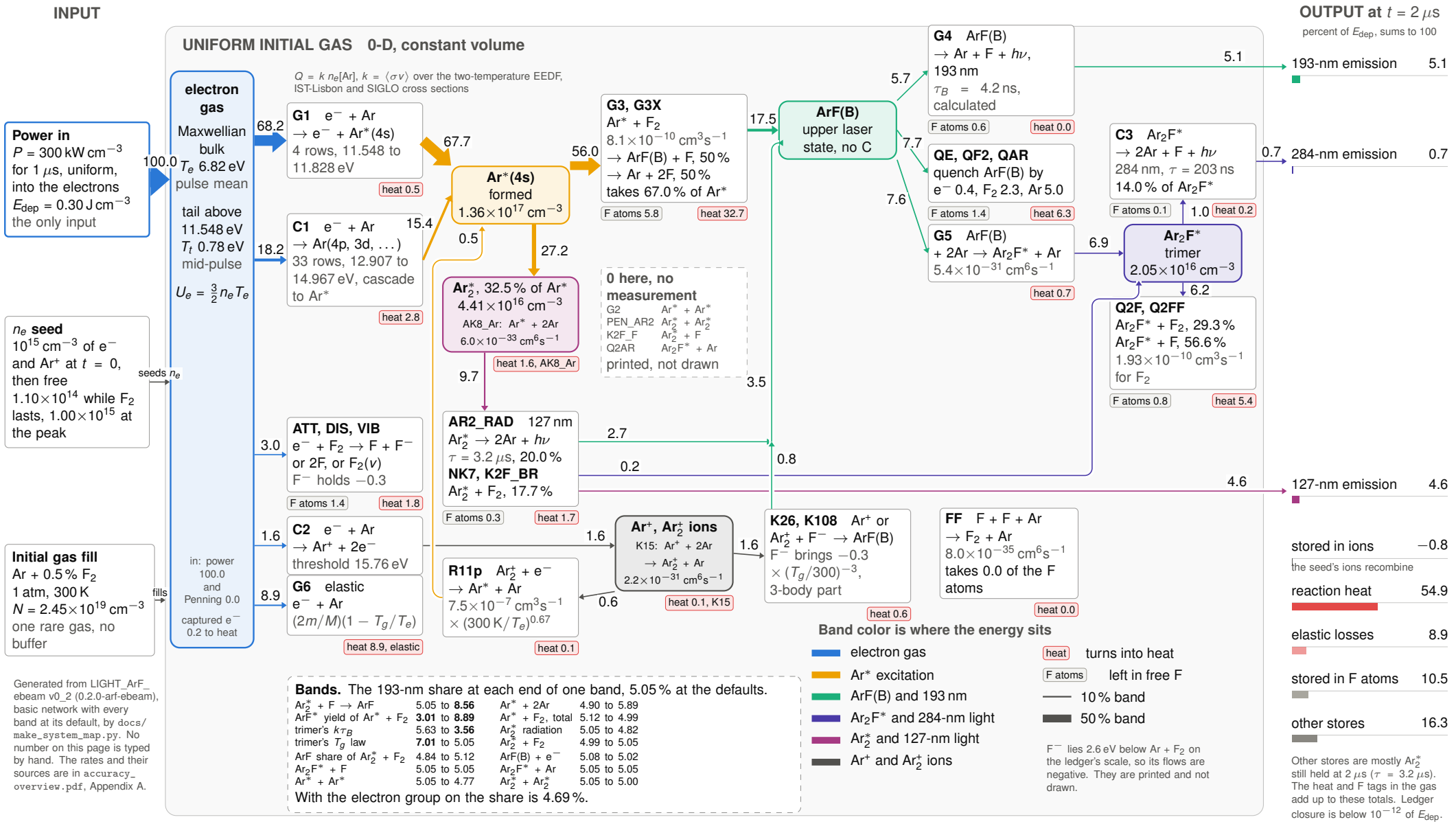


Figure 8: The energy ladder of LIGHT_ArF_ebeam v0_2, version 0.2.0-arf-ebeam, in eV above ground-state Ar and F_2 at rest. A bar is a species at its formation energy and a dashed bar a pair, the sum of two. An arrow is a row, labelled with its id. It is solid in the default network and dotted for a band whose default is 0, G2, Q2AR, K2F_F and PEN_AR2. The rows of the optional group `electron` are dashed. A photon's arrow drops by its quantum, and the red squiggle below it is the heat its lower state gives the gas as it falls apart. A red stub is a quench. The energy a row releases heats the gas, except a photon's, which leaves, and that of G2, PEN_AR2 and SUPER, which goes to the electrons. The 4 4s rows and the 33 higher rows are each drawn as one arrow to their highest threshold, and the squiggle beside each is the heat of the levels above Ar*. The rows G6, VIB1, VIB2, VIB3 and VIB4 move nothing between levels and are not drawn.

LIGHT_ArF_ebeam system map, v0_2 on LIGHT's own input: energy flow through the Ar/F_2 kinetics

Reference point: 1 atm of Ar with 0.5 % F_2 at 300 K, P = 300 kW cm^{-3} for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e seeded at 10^{15} cm^{-3} and then free, every band at its default, run to 2 μs. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1.



The notes' outputs as flows, which overlap: to Ar* 86.4 %, to Ar+ 1.6 %, to ArF(B) 18.7 %, heat 54.9 %, elastic losses 8.9 %, emission 10.3 %. Of the Ar* formed, 32.5 % forms Ar_2^* (8.7 % of the Ar* used before the F_2 burns out), and 20.0 % of that Ar_2^* radiates at 127 nm. Of the ions that recombine, 28.6 % meet an electron and return to Ar*, and 71.4 % meet F^- and form ArF(B). Of the ArF(B) formed, 5.46×10^{16} cm^{-3}, 83.2 % came from Ar* + F_2, 3.7 % from ion-ion recombination and 13.1 % from Ar_2^* + F_2. Of it 27.0 % radiates at 193 nm, 36.3 % forms the trimer, and electrons, F_2 and Ar quench 1.9 %, 11.1 % and 23.7 %. At the end of the run T_g = 675 K and p = 2.26 atm, and F_2 fell below a tenth of the fill at 728 ns.

A The reaction rates

A.1 The basic network, 69 rows

Rates at 300 K. A band row runs its band's default. N is the gas density in amagat.

row	reaction	rate	depends on	source
G1_3P2 to G1_1P1	$e^- + Ar \rightarrow e^- + Ar^*$, the four 4s levels, 11.548 to 11.828 eV	$\langle \sigma v \rangle$	T_e, T_{tail}	IST-Lisbon set, Yanguas-Gil, Cotrino and Alves 2005
C1_4p01 to C1_4dp	$e^- + Ar \rightarrow e^- + Ar(4p, 3d, 5p, 5s, 4d, 6s) \rightarrow Ar^*$, 33 levels, 12.907 to 14.967 eV	$\langle \sigma v \rangle$	T_e, T_{tail}	IST-Lisbon set, prompt cascade to 4s after Setser 1977
C2	$e^- + Ar \rightarrow 2e^- + Ar^+$	$\langle \sigma v \rangle$	T_e, T_{tail}	IST-Lisbon set, Rapp and Englander-Golden
G6	e^- + Ar, elastic	energy loss	T_e, T_g	IST-Lisbon momentum transfer, after Phelps
ATT	$e^- + F_2 \rightarrow F^- + F$	$\langle \sigma v \rangle$	T_e	SIGLO
DIS_a3	$e^- + F_2 \rightarrow e^- + 2F$	$\langle \sigma v \rangle$	T_e	SIGLO
DIS_A1	$e^- + F_2 \rightarrow e^- + 2F$	$\langle \sigma v \rangle$	T_e	SIGLO
VIB1 to VIB4	$e^- + F_2 \rightarrow e^- + F_2(v = 1$ to 4)	$\langle \sigma v \rangle$	T_e	SIGLO
G2	$2Ar^* \rightarrow Ar^+ + Ar + e^-$	0, band ar_pen		no measurement, band top from Lee and Lieberman
G3	$Ar^* + F_2 \rightarrow ArF(B) + F$	4.0×10^{-10} cm^3s^{-1}, band ar_f2 $\times$ arf_yield		total the mean of Velazco, Chen and Ueda, ArF^* yield 0.5 measured by Setser 1977
G3X	$Ar^* + F_2 \rightarrow Ar + 2F$	4.0×10^{-10} cm^3s^{-1}, band ar_f2 $\times$ (1 $-$ arf_yield)		the rest of G3, products assigned
AK8_Ar	$Ar^* + 2Ar \rightarrow Ar_2^* + Ar$	6.0×10^{-33} cm^6s^{-1}, band ar2_form		Ueda 1984, band Keto 1974 to Kolts and Setser 1977
G4	$ArF(B) \rightarrow Ar + F + h\nu$	$1/\tau_B$, τ_B 4.2 ns		τ_B calculated by Dunning and Hay, band peak Chen and Payne
QF2	$ArF(B) + F_2 \rightarrow Ar + 3F$	$k\tau_B$ 7.55×10^{-18} cm^3, over τ_B		mean of Rokni 1977 and Chen and Payne 1979
QAR	$ArF(B) + Ar \rightarrow 2Ar + F$	$k\tau_B$ 3.60×10^{-20} cm^3, over τ_B		Rokni 1977
G5	$ArF(B) + 2Ar \rightarrow Ar_2F^* + Ar$	$k\tau_B$ 2.27×10^{-39} cm^6, band arf_trimer, over τ_B		mean of Rokni, Chen and Payne, and Jancaitis, top Marowsky
QE	$ArF(B) + e^- \rightarrow Ar + F + e^-$	$k\tau_B$ 5.75×10^{-16} cm^3, band arf_e, over τ_B		Duzy 1980
C3	$Ar_2F^* \rightarrow 2Ar + F + h\nu$	4.92×10^6 s^{-1}		mean of Chen and Payne, Jancaitis, and Ueda
Q2F	$Ar_2F^* + F_2 \rightarrow 2Ar + F + F_2$	1.93×10^{-10} cm^3s^{-1}		mean of Chen and Payne, Jancaitis, and Ueda
Q2FF	$Ar_2F^* + F \rightarrow 2Ar + 2F$	1.4×10^{-10} cm^3s^{-1}, band ar2f_f		0.5 to 1 times Q2F, Jancaitis 1983
Q2AR	$Ar_2F^* + Ar \rightarrow 3Ar + F$	0, band ar2f_ar		below 10^{-14}, Jancaitis, top Boewering
K15	$Ar^+ + 2Ar \rightarrow Ar_2^+ + Ar$	2.20×10^{-31} cm^6s^{-1}	$(T_g/300)^{-0.75}$	median of eight, Brau's Table 4.8, T_g law Perez-Rios and Greene
R11p	$e^- + Ar_2^+ \rightarrow Ar^* + Ar$	7.5×10^{-7} $(T_e/300\,K)^{-0.67}$ cm^3s^{-1}	$T_e^{-0.67}$	Oskam and Mittelstadt, Mehr and Biondi, via Brau

row	reaction	rate	depends on	source
K26	$Ar^+ + F^- \rightarrow ArF(B)$	$9.32\times10^{-6}\ N/(N^2 + 0.0621N + 1.69)$ cm^3s^{-1}, N in amagat	N, screening, $(T_g/300)^{-3}$ on the 3-body limit	Hoffman and Moreno's calculation as Lee 1989 prints it
K108	$Ar_2^+ + F^- \rightarrow ArF(B) + Ar$	$1.51\times10^{-5}\ N/(N^2 + 1.81N + 2.25)$ cm^3s^{-1}, N in amagat	N, screening, $(T_g/300)^{-3}$ on the 3-body limit	the same form as K26
FF	$2F + Ar \rightarrow F_2 + Ar$	8.00×10^{-35} cm^6s^{-1}		Ganguli and Kaufman via Baulch
AR2_RAD	$Ar_2^* \rightarrow 2Ar + h\nu$	3.1×10^5 s^{-1}, band ar2_rad		Keto, Gleason and Walters 1974, top from Kannari
NK7	$Ar_2^* + F_2 \rightarrow ArF(B) + Ar + F$	4.7×10^{-10} cm^3s^{-1}, band ar2_f2 × ar2_f2_arf		Chen et al. via Chen and Payne, share Setser over Chen and Payne
K2F_BR	$Ar_2^* + F_2 \rightarrow Ar_2F^* + F$	4.7×10^{-11} cm^3s^{-1}, band ar2_f2 × (1 − ar2_f2_arf)		the rest of NK7
K2F_F	$Ar_2^* + F \rightarrow ArF(B) + Ar$	0, band ar2_f		no measurement, band top from SRI
PEN_AR2	$2Ar_2^* \rightarrow Ar_2^+ + 2Ar + e^-$	0, band ar2_pen		no measurement, band top from Lorents and Olson

A.2 The optional electron group, 5 rows, off by default

row	reaction	rate	depends on	source
KSTEP	$e^- + Ar^* \rightarrow Ar^+ + 2e^-$	$1.56\times10^{-7}\ T_e^{0.71}\ e^{-4.212/T_e}$ cm^3s^{-1}	T_e	REBUILD's template, an estimate
SUPER_3P2	$e^- + Ar^* \rightarrow e^- + Ar$	detailed balance, g ratio 1/12	T_e	detailed balance of the G1 rows, an estimate
SUPER_3P1	$e^- + Ar^* \rightarrow e^- + Ar$	detailed balance, g ratio 1/12	T_e	detailed balance of the G1 rows, an estimate
SUPER_3P0	$e^- + Ar^* \rightarrow e^- + Ar$	detailed balance, g ratio 1/12	T_e	detailed balance of the G1 rows, an estimate
SUPER_1P1	$e^- + Ar^* \rightarrow e^- + Ar$	detailed balance, g ratio 1/12	T_e	detailed balance of the G1 rows, an estimate

A.3 The bands

band	rate	default	range	unit	source
ar_f2	$Ar^* + F_2$, total	8.10×10^{-10}	6.40×10^{-10} to 1.10×10^{-9}	cm^3s^{-1}	mean of Velazco, Chen and Ueda, ends Ueda 1984 and 1983
arf_yield	ArF^* yield of $Ar^* + F_2$	0.50	0.25 to 1.00		Setser 1977, bottom Chen and Payne, top the published models
ar2_form	$Ar^* + 2Ar \rightarrow Ar_2^* + Ar$ at 300 K	6.00×10^{-33}	2.80×10^{-33} to 3.30×10^{-32}	cm^6s^{-1}	Ueda 1984, ends Keto 1974 and Kolts and Setser 1977
ar2_rad	$Ar_2^* \rightarrow 2Ar + h\nu$	3.12×10^5	3.12×10^5 to 6.00×10^7	s^{-1}	Keto 1974, top Kannari's mixing
ar2_f2	$Ar_2^* + F_2$, both channels	5.20×10^{-10}	2.50×10^{-10} to 5.20×10^{-10}	cm^3s^{-1}	Chen et al., bottom SRI
ar2_f2_arf	ArF^* share of $Ar_2^* + F_2$	0.91	0.62 to 1.00		Setser over Chen and Payne

band	rate	default	range	unit	source
ar2_f	$Ar_2^* + F \rightarrow ArF(B) + Ar$	0	0 to 1.00×10^{-9}	cm^3s^{-1}	no measurement, top SRI
ar2_pen	$Ar_2^* + Ar_2^* \rightarrow Ar_2^+ + 2Ar + e$	0	0 to 5.00×10^{-10}	cm^3s^{-1}	no measurement, top Kannari
ar_pen	$Ar^* + Ar^* \rightarrow Ar^+ + Ar + e$	0	0 to 6.20×10^{-10}	cm^3s^{-1}	no measurement, top Lee and Lieberman
arf_trimer	$ArF^* + 2Ar \rightarrow Ar_2F^* + Ar$, $k\tau_B$	2.27×10^{-39}	1.60×10^{-39} to 5.00×10^{-39}	cm^6	mean of Rokni, Chen and Payne, and Jancaitis, top Marowsky
trimer_tg_pow	$ArF^* + 2Ar \rightarrow Ar_2F^* + Ar$, its $(T_g/300)^n$	0.00	−3.90 to 0.00		no measurement, bottom NRL's form after Shui
arf_e	$ArF^* + e$, $k_e\tau_B$	5.75×10^{-16}	4.00×10^{-16} to 7.50×10^{-16}	cm^3	Duzy 1980, two measurements
ar2f_f	$Ar_2F^* + F \rightarrow 2Ar + 2F$	1.45×10^{-10}	9.65×10^{-11} to 1.93×10^{-10}	cm^3s^{-1}	Jancaitis 1983
ar2f_ar	$Ar_2F^* + Ar$	0	0 to 2.20×10^{-14}	cm^3s^{-1}	below 10^{-14}, Jancaitis and Marowsky, top Boewering

A.4 Formation energies

species	eV above Ar + F_2	how known	source
Ar	0.000		ground state
F_2	0.000		ground state
F	0.800	measured	half the F_2 bond, REBUILD
F^-	−2.600	measured	F less the electron affinity, REBUILD
Ar^*	11.548	threshold	IST-Lisbon 3P2 threshold
Ar^+	15.760	threshold	IST-Lisbon ionization threshold
Ar_2^+	14.530	measured	well depth by photoionization, via Krauss and Mies
Ar_2^*	10.868	calculated	well depth of Saxon and Liu
ArF(B)	7.214	lower bound	F plus the 1933-Å band peak of Chen and Payne
Ar_2F^*	6.564	calculated	well depth of Huestis, Marowsky and Tittel

B The data scored

The 16 figure readings in `arf_data/fig_readings/`, each with its source and the way it was read. The numbers printed in the sources' text are in `arf_data/SCORING_DATA.md`.

file	source	reading
`arf_device_gain_points`	ArF (Ar/F2) small-signal gain points as plotted by NRL and Keio, survey readings at 300 dpi (colour-marker centroids), P = e-beam power density	
`chenpayne1979_fig13_ArF_decay`	Chen and Payne, IEEE JQE 15, 149 (1979), Fig. 13 p.153: ArF (193.3 nm) decay frequency vs F2, proton excitation	survey reading of the PDF at 300 dpi, ±0.1e7 per s
`chenpayne1979_fig14_Ar2F_decay`	Chen and Payne, IEEE JQE 15, 149 (1979), Fig. 14 p.154: Ar2F (284 nm) decay frequency vs F2, P_Ar 300 and 600 Torr (symbols not separated)	survey reading of the PDF at 300 dpi, ±0.05e7 per s
`chenpayne1979_fig16_17_SternVolmer`	Chen and Payne, IEEE JQE 15, 149 (1979) p.155. Fig.16: 1/I(193.3) vs F2 at P_Ar 600 Torr. Fig.17: 1/I(193.3) vs P_Ar^2 (1e5 Torr^2) at P_F2 0.84 Torr	survey reading of the PDF at 300 dpi

file	source	reading
electra_petrova2022_fig3_pump	Petrova et al., NRL/6720/MR–2022/1, Fig. 3 printed p.9 (PDF 16): measured single-shot Electra e-beam power density vs time	survey trace of the PDF at 300 dpi, ±5 kW/cm3, integral 67.9 mJ/cm3 (trapezoid, survey arithmetic)
jancaitis1983_fig3_peak_vs_current	Jancaitis and Powell, UCRL-53465 (1983) App. Ch. V, Fig. 3 printed p.165 (PDF 173): peak ArF and Ar2F fluorescence (arb., log) vs average current density, 5 Torr F2 - 1500 Torr Ar, 600 keV, 50 ns	survey reading of the PDF at 200 dpi, ±8 percent on both axes
jancaitis1983_fig7_Ar2F_decay	Jancaitis and Powell, UCRL-53465 (1983) App. Ch. V, Fig. 7 printed p.169 (PDF 177): Ar2F decay rate vs total pressure, (a) Ar/F2 = 300, (b) 600, 11 and 33 A/cm2 not separated	survey reading of the PDF at 300 dpi, ±0.07e7 per s
morozov2008_fig10_Ar_efficiency	Morozov et al., J. Appl. Phys. 103, 103301 (2008), Fig. 10 Ar panel p.103301-7: VUV efficiency (4pi emitted over deposited) vs pressure, two symbol sets not separated	survey reading of the PDF at 300 dpi, systematic ±0.04 (paper)
rokni1977_PRA_fig3_F2	Rokni, Jacob and Mangano, Phys. Rev. A 16, 2216 (1977), Fig. 3 p.2219: 1/I(ArF) vs F2, P_Ar 150 Torr, 5 A/cm2, 150 keV, 300 ns	survey reading of the PDF at 300 dpi, bottom y label printed 0.8 read as 0.08, ±0.004
rokni1977_PRA_fig4_Ar	Rokni et al. PRA 16, 2216 (1977), Fig. 4 p.2219: ArF fluorescence (relative) vs Ar, 2 Torr F2, 5 A/cm2, not normalised to deposition	survey reading of the PDF at 300 dpi, ±2 units
rokni1977_PRA_fig5_current_ratio	Rokni et al. PRA 16, 2216 (1977), Fig. 5 p.2219: I(high)/I(low) for a x25 e-beam current change, 2 Torr F2	survey reading of the PDF at 300 dpi, ±0.15
sri1976_fig11_ArF_yield_vs_F2	Nakano et al., SRI MP-76-99 (1976), Fig. 11 printed p.52 (PDF 56): ArF (193 nm) yield vs F2, Kr 0, Ar 1, 2, 5 atm, log axis	survey reading of the PDF at 200 dpi, ±8 percent
sri1976_fig4_yields_vs_Ar	Nakano et al., SRI MP-76-99 (1976), Fig. 4 printed p.26 (PDF 31): fluorescence yields (photons per initially produced excited state) vs Ar, F2 1 Torr, Kr 0, Febetron 706	survey reading of the PDF at 250 dpi, ±0.01
ueda1983_fig10_ArMeta_pureAr	Ueda et al., Rev. Laser Eng. 11, 576 (1983), Fig. 10 p.38 (583): Ar(3P2) 696.5-nm absorption decay rate in pure Ar vs pressure (p^2 axis), two probe widths	survey reading of the PDF at 300 dpi, ±0.1e7 per s
ueda1983_fig12_ArMeta_vs_F2	Ueda et al., Rev. Laser Eng. 11, 576 (1983), Fig. 12 p.584 (footer 39, PDF 9): Ar(3P2) absorption decay rate vs F2 fraction, Ar-F2 at 2.5 atm, printed k = 1.1e-9 cm3/s	survey reading of the PDF at 300 dpi, ±0.3e7 per s (error bars in the figure are about ±0.7e7)
ueda1983_fig8_Ar2F_decay	Ueda, Kanada, Kitagawa and Takuma, Rev. Laser Eng. 11, 576 (1983), Fig. 8 p.36 (581): Ar2F decay rate vs F2 density, Febetron 706, 3 ns, 500 kV	survey reading of the PDF at 300 dpi, ±0.1e7 per s